

Construction Cost Minimization of Shallow Foundation Using Recent Swarm Intelligence Techniques

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Abstract—In this study, the performances of eight recent swarm intelligence techniques, accelerated particle swarm optimization (APSO), firefly algorithm, levy-flight krill herd, whale optimization algorithm (WOA), ant lion optimizer, grey wolf optimizer, moth-flame optimization algorithm and teaching-learning-based optimization algorithm (TLBO), are explored. Particle swarm optimization algorithm is also considered to benchmark the efficiencies. A final cost is considered as an objective function which deals with shallow footing optimization with two attitudes: routine optimization, and sensitivity analysis. Moreover, as a further study, the effect of the location of the column at the top of the foundation is examined by adding two spare design variables. To this end, three numerical case studies are simulated. Based on the final results TLBO showed an acceptable performance because of the lowest mean values and WOA demonstrated the weakest efficiency among the algorithms in this study.

Index Terms—Construction industry, global optimization, shallow footing, swarm intelligence techniques.

I. INTRODUCTION

SPREAD footing belonging to the geotechnical engineering as one of the most important and sensitive structural elements attracts wide attention in the recent investigations. Structures functioning can be ruined unless a well-designed foundation directs the effective loads to the earth successfully. Therefore, the true design of shallow footings takes much attention in the field of civil engineering.

In the past few decades, metaheuristic optimization algorithms as effective alternative facilitate complex problems very well [1]. Recently, optimization algorithms have been widely used in facilitating construction industry related problems, e.g.,

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structural engineering [2]–[5], water engineering, geotechnical engineering, and transportation engineering [6]–[10]. Although the literature on different subjects of civil engineering optimization is extensive, there are very limited studies on the basis of the optimum design of shallow footings. According to Khjaehzadeh *et al.* [11], a hybrid approach founded on combining the firefly algorithm (FA) with the sequential quadratic programming, namely FaSqp, utilized for the optimum design of shallow footings.

Besides, Camp and Assadollahi [12] took two different objectives, minimum-cost, and minimum-Co2 emission, interactions into account in the case of uniaxial loading case by applying Big Bang-Big Crunch. Then, Camp and Assadollahi [13] considered cost and CO₂ emission design of footing under the uniaxial uplifting case.

The present study is examining the performance of eight recently developed swarm intelligence techniques, the whale optimization algorithm (WOA); levy flight krill herd (LKH); FA; accelerated particle swarm optimization algorithm (APSO), ant lion optimizer (ALO), grey wolf optimizer (GWO), moth-flame optimization algorithm (MFO), and teaching-learning-based optimization algorithm (TLBO) for the optimum design of shallow footing problems for the first time. Successful application of the mentioned optimization algorithms on civil engineering problems demonstrates their ability for handling here proposed issue. To this end, a MATLAB code is developed based on ACI 318-05 requirements [14]. In this way, the objective function is considered as the total cost of a shallow footing construction. Two different loading cases of the uniaxial and flexural moment are considered for numerical simulation presented here. Furthermore, as a further investigation, the effect of changing the column location on the footing surface is taken into account in the third case study. This paper also explores the impact of different input parameters on the final design by providing a sensitivity analysis on friction angle, the elasticity of modulus, Poisson's ratio, and density of the base soil, in addition to the depth of the footing and inclination of the effective load respect to the vertical direction.

II. METHODOLOGY

Fig. 1(a) shows a schematic view of a shallow footing. A footing can be expressed by L , B , and H which represent length, width, and thickness of footing, respectively. Other important

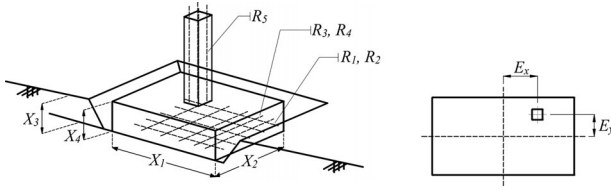


Fig. 1. Schematic view and design variables for describing a shallow footing.

parameters for describing foundations are the depth of the bottom of footing from the ground surface (D), the column width (b_{col}), the over-excavation length and width (L_0 and B_0).

The final design has to be checked for geotechnical stability and structural strength. Geotechnical stability measures are bearing capacity [15] and settlement [16].

In addition to the geotechnical criteria, numbers of structural standards based on ACI 318-05 [14] (i.e., one-way shear strength, two-way shear strength, flexural strength, and development lengths) have to be considered in the final design. The detail of utilized limitations and constraints can be found in ACI 318-05 [14] and Camp and Assadollahi [13].

III. OBJECTIVE FUNCTION FOR OPTIMIZATION

Refer to Fig. 1(a), in order to model a shallow footing, nine design variables is defined accordingly: the length (X_1), the width (X_2), the thickness (X_3); the depth of footing (X_4), the bar number in the long direction (R_1), the number of bars in the long direction (R_2), the bar number in short direction (R_3), the number of bars in short direction (R_4), and bar number of the dowels (R_5). As an additional investigation two spare design variables, E_x and E_y , are defined for locating the column at the top of the foundation as shown in Fig. 1(b).

The objective function for minimum cost design of shallow footing is defined as

$$f_{\text{cost}} = C_e V_e + C_f A_f + \xi C_r M_r + \frac{f'_c}{f'_{c \min}} C_c V_c + C_b V_b \quad (1)$$

where C_e is the unit cost of excavation, C_f is the unit cost of the framework, C_r is the unit cost of reinforcement, C_c is the unit cost of concrete, C_b is the unit cost of backfill, respectively.

IV. OPTIMIZATION ALGORITHMS DESCRIPTION

In the past few decades, metaheuristic optimization algorithms by imitating nature's disciplines attitude the complicated problems in different fields very well [17]–[19]. A short description of utilized algorithms in the current study is presented accordingly.

A. Accelerated Particle Swarm Optimization Algorithm

Recently, Yang [18] proposed a new variant of PSO called APSO by considering two main attitudes; faster convergence and simpler application than the original PSO algorithm. In order to provide faster convergence, the particles' best position term is removed from the original velocity formulation.

In lieu, APSO considers ε_n , a random number with normal distribution within $[0,1]$, as a substitution for this removed term from the velocity equation.

B. Firefly Algorithm

FA models the behavior of fireflies for finding foods and mates mathematically [19]. In this way, all fireflies prefer to move toward brighter positions.

FA works based on three fundamental rules accordingly: 1-Consider all the fireflies unisex which are inclined to converge toward brighter one regardless of their sex. 2- Attractiveness is related to brightness and the inverse of the distance from other fireflies. In the case of lack of brighter individual, each firefly will move randomly. 3- The brightness of fireflies is defined based on the objective function values. The fitter the solution, the brighter the firefly would be.

C. Levy Flight Krill Herd

Recently, krill response to the biological and environmental stimulations draws the attention of Gandomi and Alavi [20] to develop krill herd (KH) algorithm. KH is one of the swarm intelligence techniques which modeled krill behavior for finding food supply. The herding of the krill tracks two basic goals: increasing krill density and reaching food.

In an attempt, Wang *et al.* [21] utilized levy flight as a local search engine for improving the exploitation ability while promising characteristics of the KH are preserved. LKH use the usual procedure of standard KH for contracting the solution space before applying local levy flight for exploiting solution.

D. Whale Optimization Algorithm

Humpback whales' fascinating mechanism of hunting enticed Mirjalili and Lewis [22] to introduce the WOA. To be more exact, the humpback whales utilized an exploration procedure called bubble-net feeding method [38]. Humpback whales desire to hunt aschool of krill or small fishes near the surface. This foraging is accomplished by creating distinctive bubbles along a circle or '9'-shaped path. In order to do them two different tactics will be utilized: upward-spirals and double-loops.

Mathematical modeling of this algorithm considers the following two kinds of movements for the humpback whales accordingly:

- 1) shrinking encircling mechanism;
- 2) spiral updating position where guide whales' movement within a spiral-shaped path by considering the distance between the whale and prey locations.

E. Ant Lion Optimization Algorithm

ALO algorithm is one the most recent bioinspired optimization techniques developed by Mirjalili [23]. This algorithm has modeled interesting behavior of antlions (doodlebugs) for trapping their prey. Generally, antlions life cycles are consisting of two major periods; larvae and mature. ALO imitates the foraging behavior of antlions' larvae mainly. To be more exact, ALO remodels the antlions dealings with ants. Basically, as ants

TABLE I
DESIGN COST VALUES FOR THE NUMERICAL CASE STUDIES

Optimization Algorithm	PSO	APSO	FA	LKH	WOA	ALO	GWO	MFO	TLBO
Case I									
Best	43,442.27	43,443.57	43,460.08	43,442.27	44,069.07	43,449.55	43,449.51	43,442.27	43,442.27
Worst	174,424.19	101,988.36	56,392.56	121,053.07	269,536.09	86,639	50,867.28	92,818.80	80,530.39
Mean	68,735.94	59,688.07	47,889.69	60,307.73	74,438.94	54,489.32	44,099.64	52,244.01	50,030.90
SD	25,816.84	13,129.36	4,050.38	20,401.16	33,273.80	9,100.66	1,896.90	11,878.01	6,983.25
Case II									
Best	N.G.	83,106.50	83,420.23	82,561.80	86,211.90	82,591.50	82,913.69	82,561.80	82,561.80
Worst	N.G.	199,333.61	95,344.62	168,203.28	213,337.85	171,792.51	86,744.27	193,416.99	91,131.93
Mean	N.G.	111,235.31	87,260.74	103,150.55	143,928.04	91,960.91	84,424.89	98,280.99	84,042.91
SD	N.G.	39,132.04	2,048.08	31,489.16	28,789.61	11,750.99	905.71	17,005.85	1,319.29
Case III									
Best	50,774.31	50,477.45	50,582.63	48,720.24	62,785.04	52,372.17	49,218.44	49,141.25	49,504.24
Worst	133,384.03	153,021.38	71,602.48	625,896.24	726,015.73	89,347.06	113,860.89	220,477.60	80,344.96
Mean	60,345.81	65,808.17	59,828.52	75,764.29	147,202.86	64,734.54	62,590.00	84,223.73	56,061.12
SD	14,491.96	14,359.98	3,806.46	60,913.08	96,514.92	7458.81	21403.45	34322.99	5078.1

TABLE II
FINAL LOW COST OPTIMIZATION DESIGNS FOR THE PROPOSED CASE STUDIES

Optimization Algorithms	X1 (m)	X2 (m)	X3 (m)	X4 (m)	R1	R2	R3	R4	R5	Ex	Ey
Case I											
PSO	1523.84	1523.84	1030.40	300	16	11	16	11	14	–	–
APSO	1523.84	1523.84	1040.28	300.02	16	11	16	11	14	–	–
FA	1524.22	1523.86	1132.76	300	16	11	16	11	14	–	–
LKH	1523.84	1523.84	1030.40	300	16	11	16	11	14	–	–
WOA	1883.92	1695.57	1370.00	348.01	18	10	18	11	14	–	–
ALO	1523.84	1523.84	1065.78	303.00	16	11	16	11	14	–	–
GWO	1523.96	1523.97	1060.02	300.70	16	11	16	11	14	–	–
MFO	1523.84	1523.84	1030.40	300	16	11	16	11	14	–	–
TLBO	1523.84	1523.84	1030.40	300	16	11	16	11	14	–	–
Case II											
PSO	–	–	–	–	–	–	–	–	–	–	–
APSO	1695.57	2900.00	1224.88	303.47	18	17	18	9	14	–	–
FA	1695.58	2900.70	1221.04	308.37	18	17	14	15	14	–	–
LKH	1695.57	2900.00	1035.19	334.76	18	16	16	12	14	–	–
WOA	1735.47	2900.00	2762.60	335.87	18	16	14	16	16	–	–
ALO	1695.57	2900.00	1162.46	335.57	18	16	16	12	14	–	–
GWO	1695.68	2900.06	2589.86	335.32	18	16	16	12	14	–	–
MFO	1695.57	2900.00	1035.19	334.76	18	16	16	12	14	–	–
TLBO	1695.57	2900.00	1035.19	334.76	18	16	16	12	14	–	–
Case III											
PSO	1684.01	1739.08	1798.74	300.00	16	12	14	15	14	0.00	–193.49
APSO	1909.81	2414.18	1976.07	353.06	18	13	20	9	14	62.83	–58.78
FA	1502.06	1713.47	2089.90	347.33	14	16	14	16	14	9.46	–179.42
LKH	1382.48	1726.56	2075.17	383.65	14	17	14	15	14	–0.64	–187.22
WOA	2625.68	4000	864.70	358.47	20	18	16	18	18	43.18	492.14
ALO	1695.57	1739.08	2861.71	300.00	18	10	14	15	14	–0.00	–193.49
GWO	1411.85	1739.96	2173.91	379.18	14	17	14	15	14	–8.49	–193.71
MFO	1408.55	1739.08	2008.21	379.66	14	17	14	15	14	–0.00	–193.49
TLBO	1541.67	1739.08	1903.64	339.68	14	17	14	15	14	0.00	–193.49

exploring the area by go through the earth surface, they are highly likely to be trapped by predators.

F. Grey Wolf Optimizer

GWO introduced by Mirjalili *et al.* [24] based on emulating hunting technique and the social hierarchy of grey wolves. Mostly, they live in a group with a size of 5–12 members.

Individuals can be categorized into four main classes with specific responsibility proportional to their abilities. At the top of the group is the leader named alpha make important decision essential for group survival like hunting, sleeping place, time to wake, and so on. Alpha’s associate called beta also plays an important role in managing the group in case of alpha absence. The lowest social class yet vital for the group solidarity is omega acts as a scapegoat. In fact, they always have to submit to all

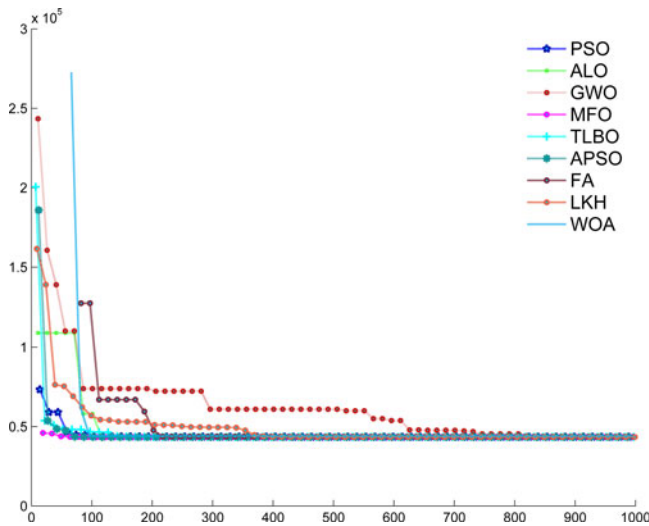


Fig. 2. Convergence rate plot for the best-found solutions for case I.

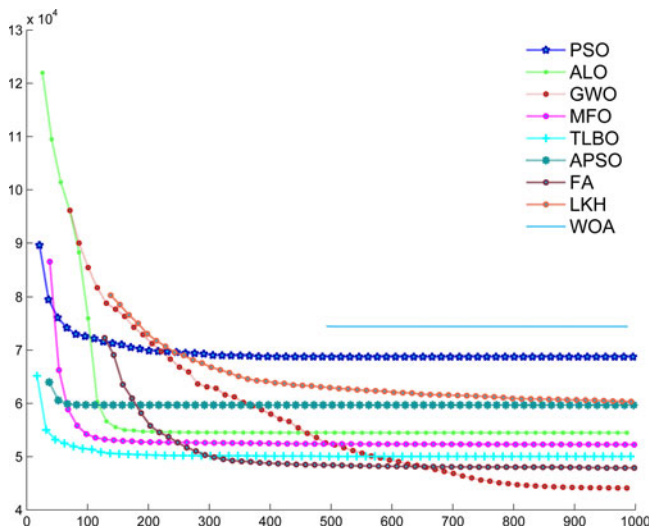


Fig. 3. Convergence rate plot for mean of the solutions for case I.

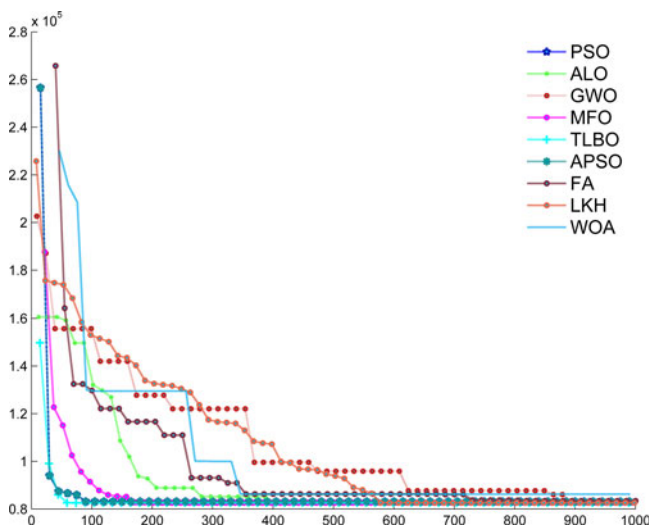


Fig. 4. Convergence rate plot for the best-found solutions for case II.

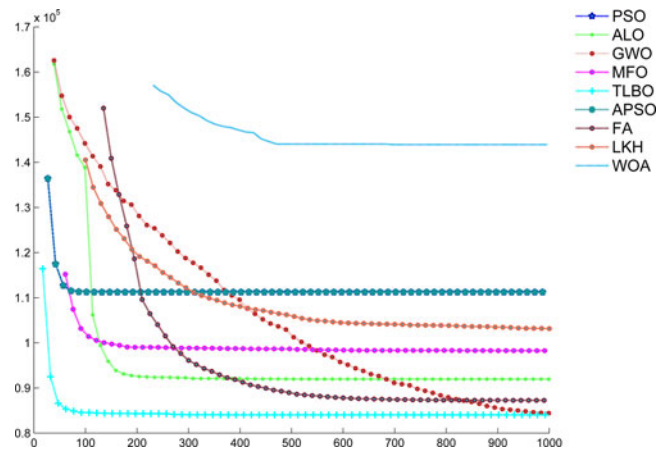


Fig. 5. Convergence rate plot for mean of the solutions for case II.

the other dominant parts. Finally, the delta wolves are the ones who are in three mentioned categories. They are subordinate for both alpha and beta and above the omegas.

Main characteristics of grey wolf hunting can be summarized as follows based on Muro *et al.* [25]:

- 1) Tracking, chasing, and approaching the prey.
- 2) Pursuing, encircling, and harassing the prey until it stops moving.
- 3) Attack towards the prey.

G. Moth Flame Optimization

MFO proposes by Mirjalili [26] mimics the navigating behavior of moths in the night. Moths try to keep moving by a fix inclination with respect to month, a method named transverse orientation for navigation. However, the light supply has to be far enough and they can be tricked by some artificial light sources that are close them. In the case of too close light supply, it may be seen that moths eventually converge toward the light. MFO collect this reaction in order to find the global optimum solution based on mathematical modeling.

In this algorithm, thespiral function is considered for modeling the path moths follow to reach the flame. The utilized spiral has to have three features as follows: 1- Spiral's initial point should start from the moth. 2- Spiral's final point should be the position of the flame. 3- Fluctuation of the range of spiral should not exceed from the search space.

H. Teaching-Learning Based Optimization

TLBO algorithm accounts the difference between the performances of two different teachers for transferring a specific subject to the similar learners [27], [28]. In this algorithm two mechanisms are introduced for the learning: 1- through teacher (teacher phase) 2- through interaction with the other learners (learner phase). The basic analogies between the notions in this method and optimization procedure are group of learners as population, different subjects offered to learners as design variables and learners' outputs as fitness values.

TABLE III

COST (\$) DESIGN VARIATION UNDER VARYING DEPTH OF FOUNDATION, COMPRESSIVE STRENGTH OF CONCRETE, AND EFFECTIVE FORCE INCLINATION RESPECT TO THE VERTICAL DIRECTION FOR CASE II

		PSO	APSO	FA	LKH	WOA	ALO	GWO	MFO	TLBO
D_f (m)	300	–	133343.8	134297.32	133257.41	134474.42	133256.7	133257.7	133282.3	133315.09
	500	–	114248.48	114703.17	114247.24	116723.12	113720.4	113719.1	114294.2	113718.70
	700	–	96200.16	99215.71	95742.40	100893.19	96405.16	95718.41	96523.15	95741.71
	800	–	89485.58	90737.22	89088.78	92097.86	89485.58	89091.97	89088.78	89088.78
	1000	–	80630.96	81113.44	79848.58	84057.32	79848.58	79860.27	79961.05	80630.96
	1200	–	73615.09	77657.68	71824.52	79277.67	74347.46	71827.09	72090.87	71824.52
	1400	–	71448.91	71496.13	71226.60	75222.72	71245.73	71240.98	71226.6	71226.69
	1600	–	71264.20	71509.25	71264.20	72962.08	71264.21	71287.84	71264.2	71264.20
	1800	–	71319.30	71459.57	71319.30	71691.49	71301.81	71326.82	71301.8	71303.25
	2000	–	71356.09	71598.24	71339.40	73288.32	71356.09	71353.99	71339.4	71339.41
	2200	–	71376.99	71612.46	71377.00	71971.55	71392.88	71400.52	71392.88	71392.88
	2400	–	71414.59	71492.09	71414.60	72689.25	71429.67	71427.83	71414.59	71429.66
	2600	–	71452.19	71529.59	71452.19	72757.93	71452.2	71466.49	71452.19	71466.45
	2800	–	71494.24	71738.54	71706.81	72826.29	71489.81	71505.61	71489.79	71503.24
	3000	–	71527.39	71764.97	71527.39	73048.98	71527.41	71540.37	71527.39	71527.39
Variation (%)		–	46.36	46.56	46.32	45.68	46.32	46.31	46.33	46.35
f_c (MPa)	20	–	71346.59	71545.62	71230.12	73121.39	71483.42	71287.890	71233.95	71233.95
	25	–	66006.51	66749.35	66049.56	66742.11	66178.58	65982.10	65977.78	66539.661
	30	–	62533.15	62639.31	62581.04	63385.61	62232.294	62083.787	61974.73	62524.83
	35	–	59114.49	59922.06	59189.25	61695.27	59572.65	59156.26	59157.51	59249.38
	40	–	57694.51	58231.12	57108.29	58588.47	57194.24	57169.35	57145.55	57162.29
	45	–	57219.09	57429.85	55817.40	59357.76	56132.93	55931.86	56057.27	56362.74
	50	–	55408.80	57446.04	57345.84	60060.71	55329.65	55406.34	55515.36	55294.53
	55	–	56399.08	56604.97	56095.35	56887.87	55681.03	55496.40	55714.5	55740.58
Variation (%)		–	20.95	20.88	21.25	22.20	22.11	22.15	21.79	21.75
i (°)	0	–	71280.46	71531.95	80588.87	73099.69	71387.20	71278.74	71202.95	71233.95
	5	–	71287.62	71590.60	80634.69	73376.89	71298.18	71320.42	71247.23	71278.47
	10	–	71454.96	71630.40	80687.74	74361.63	71400.91	71341.98	71295.62	71295.66
	15	–	71466.23	71723.24	80745.58	73810.29	71489.26	71406.00	71348.82	71380.54
	20	–	71666.25	71730.95	80836.82	77932.19	71675.26	71474.29	71407.75	71439.72
	25	–	72415.54	74460.65	98445.36	87899.11	73145.69	71535.21	71540.03	71527.97
	30	–	75894.80	79077.67	99780.59	85687.63	79059.24	72467.95	72432.81	79056.02
	35	–	82579.05	82604.70	101020.70	89776.28	82579.05	82580.61	82579.07	82579.05
	40	–	91689.24	92917.78	123234.55	108347.12	90266.60	90272.97	90266.61	90266.60
	45	–	110455.94	116019.50	242507.45	121247.97	106376.95	103109.93	103129.82	102860.95
Variation (%)		–	–54.96	–62.19	–200.92	–65.87	–49.01	–44.66	–44.84	–44.40

V. NUMERICAL SIMULATIONS

Similar to most of the metaheuristic optimization algorithms here utilized techniques require exact setting of the internal parameters to perform in the most efficient possible way. Hence, the recommended parameter setting proposed by the authors of the original papers for all the utilized algorithms are considered in the present study. In order to assess the efficiency of the proposed algorithms, this section is devoted to simulating three case studies. In this way, for the first case, a shallow footing is considered for a uniaxial loading yield from dead and live loads of 650 kN and 350 kN, respectively. In the second case, an effective moment from the dead and live load as 400 kN.m and 150 kN.m is added to the uniaxial force of the previous case. The third case explores the impact of relocating the column at the top of footing in each direction. Due to the chaotic fundamental of the optimization algorithms a series of 101 runs are fulfilled and all the results are reported based on best, worst, mean, median, and standard deviation (SD). The whole algorithms have been run for the population size of 50 and iterations number of

1000. As a further study, a sensitivity analysis is conducted on soil parameters variation accordingly: base soil friction angle variation between 28° and 38°, soil density between 11.5 and 23.5 kN/m³, modulus of elasticity between 10 500 and 90 500 kPa, Poisson's ratio between 0.1 and 1, concrete compressive strength between 20 and 55 MPa, effective force inclination respect to vertical axis between 0° and 45°, and depth of footing between 300 and 3000 mm. The column for conveying the loading to the footing has an area of 400 × 400 mm² with reinforcements' composition of 6 Φ 16.

A. Case I: Uniaxial Loading

This case is resolved based on a routine optimization. Table I has gathered the final cost design in the form of the best, worst, mean, and SD values. The final design variables are also tabulated in Table II. In this case, PSO, LKH, MFO, and TLBO reached the lowest design of \$43,442.27. However, FA and GWO by achieving the lower value of the mean, SD, and worst performed better than other utilized here algorithms. It

TABLE IV

COST (\$) DESIGN VARIATION UNDER VARYING FRICTION ANGLE OF BASE SOIL, SOIL DENSITY, MODULUS OF ELASTICITY, AND POISON'S RATIO CASE II.

		PSO	APSO	FA	LKH	WOA	ALO	GWO	MFO	TLBO
φ (°)	28	–	71389.98	71630.36	80670.65	72771.53	71341.75	71338.56	71281.41	71313.03
	30	–	71404.80	71553.41	80619.59	72899.19	71327.39	71336.05	71233.54	71264.91
	32	–	71425.07	71450.96	80540.45	72517.51	71332.27	71247.87	71314.9	71191.65
	34	–	71185.32	71430.39	80483.26	73853.96	71320.40	71169.51	71099.70	71129.44
	36	–	71216.59	71372.67	80394.86	73740.56	71090.08	71075.06	71031.49	71060.04
	38	–	71090.85	71380.87	80357.77	72755.46	71132.42	71052.94	71003.78	71003.78
	Variation (%)	–	0.42	0.35	0.39	0.02	0.293	0.400	0.389	0.434
γ (°)	11.5	–	71436.20	71752.48	80815.99	74862.98	71441.43	71420.80	71378.65	71411.39
	13.5	–	71361.71	71701.03	80712.04	73479.59	71389.49	71395.06	71317.91	71350.16
	15.5	–	71398.22	71623.04	80661.02	73324.03	71415.87	71350.39	71505.10	71302.76
	17.5	–	71272.35	71587.43	80622.52	72240.59	71286.17	71300.44	71264.91	71264.91
	19.5	–	71319.12	71529.86	80585.73	74294.55	71384.23	71270.45	71233.95	71208.24
	21.5	–	71275.29	71486.30	80557.32	72766.60	71451.18	71237.93	71177.46	71180.03
	23.5	–	71255.42	71554.91	80533.16	74300.25	71268.47	71217.38	71155.89	71164.01
	Variation (%)	–	0.25	0.27	0.35	0.75	0.24	0.28	0.31	0.35
E_s (kPa)	10500	–	71265.45	71582.72	80589.42	73278.13	71250.95	71266.73	71233.95	71233.95
	20500	–	71402.22	71584.07	80587.59	74118.23	71433.15	71275.30	71202.95	71233.95
	30500	–	71255.08	71623.97	80585.81	74091.80	71294.79	71264.48	71436.34	71233.95
	40500	–	71297.83	71564.12	80585.71	73290.92	71418.31	71270.50	71202.95	71233.95
	50500	–	71330.17	71538.37	80586.20	73002.30	71263.64	71277.12	71202.95	71202.95
	60500	–	71356.61	71550.59	80594.18	73345.48	71239.19	71263.95	71233.95	71233.95
	70500	–	71227.00	71567.19	80586.00	71466.34	71348.31	71243.78	71233.95	71233.95
	80500	–	71313.15	71599.71	80585.93	74439.30	71397.81	71253.76	71233.95	71202.95
	90500	–	71280.84	71516.79	80585.74	74679.43	71272.78	71279.43	71233.95	71233.95
	Variation (%)	–	–0.02	0.09	0.00	–1.91	–0.03	–0.02	0	0
ν	0.1	–	71306.27	71551.80	80586.34	73615.48	71275.49	71270.60	71233.95	71233.95
	0.2	–	71316.11	71571.09	80585.71	72319.55	71341.47	71270.86	71202.95	71233.95
	0.3	–	71299.12	71584.38	80585.71	72003.10	71277.90	71262.30	71202.95	71233.95
	0.4	–	71325.98	71609.42	80585.81	73181.22	71287.88	71277.01	71202.95	71233.95
	0.5	–	71289.83	71568.23	80588.50	72786.32	71269.59	71277.04	71202.95	71233.95
	0.6	–	71277.27	71508.96	80585.71	71728.11	71284.43	71279.10	71202.95	71233.95
	0.7	–	71283.33	71518.59	80586.60	72545.20	71332.35	71253.41	71202.95	71233.95
	0.8	–	71248.77	71530.75	80585.79	72931.11	71320.37	71287.68	71202.95	71211.82
	0.9	–	71227.57	71574.86	80586.60	74187.24	71460.35	71233.67	71233.95	71233.95
	1	–	71217.54	71560.36	80585.72	73369.20	71416.54	71270.46	71202.95	71233.95
	Variation (%)	–	0.12	–0.01	0.00	0.33	–0.198	0.000	0.0435	0

can be concluded that GWO is the best algorithm in this case of study based on the lowest values for Worst, Mean, and SD of \$50867.28, \$44099.6, and \$1896.90, respectively.

Moreover, convergence rate histories of the Best and Mean results (shown in Figs. 2 and 3) clearly prove the observations presented in Table I. Fig. 2, demonstrates that PSO, MFO, and TLBO converged to the best solution from the initial iterations while LKH and GWO started converging to the final result after 400 iterations. From the Fig. 3, it can be seen that FA converged to the final result faster than GWO, though GWO with a consistent downward pattern has been more successful to find the lower value of Mean. Moreover, it is obvious that WOA fails to reach the permitted solution till about 500 iterations, but also it ended up to the highest value of Mean among here utilized algorithms.

B. Case II: Axial and Flexural Loading at the Center of Foundation Uniaxial Loading

In this case, in a series of 101 runs, the best, the worst, mean, and SD values of low-cost design are provided by Tables I and II. As demonstrated in Table I, PSO is not capable of converging to

a valid final design. The final cost design, in this case, implied that TLBO and GWO are the best algorithms because of the lower values of mean, SD, and worst than other here utilized algorithms. As mentioned in Table I, TLBO with Mean value of \$84,042.91 is better than GWO with \$84,424.89.

However, GWO with lower SD and Worst values of \$905.71, and \$86,744.27 proves more uniform performance rather than TLBO with SD and Worst values of \$1,319.29 and \$91,131.93. Moreover, in this case, LKH, MFO, and TLBO achieved the lowest value of Best of \$82,561.80. The maximum Mean value of \$143,928.04, by WOA, is about 71.25% greater than TLBO while other Mean values by APSO, FA, LKH, ALO, GWO, and MFO are about 32.35%, 3.83%, 22.73%, 9.42%, 0.45%, and 16.94%, greater than TLBO, respectively.

Convergence rate plots based on Best and Mean values also presented in Figs. 4 and 5, respectively. Best results as Fig. 4, shows that MFO, APSO, and TLBO get close to the final result faster than other algorithms from the initial iterations.

Moreover, LKH started converging to the final solution from the middle iterations while GWO did not converge to the final solution until the latest iterations. Fig. 5 visualizes the defi-

nite superiority of TLBO in converging to the final design (before 100th iteration). Moreover, GWO mean results convergence again shows the consistent and late convergence to the final results. WOA again was the weakest method among here tackled techniques. Therefore, the results propose TLBO as the most efficient algorithm. Therefore, in this case, TLBO with the lowest values of both Best and Mean performed better than other here utilized algorithms. At the next places, GWO and FA are stood with 0.45% and 3.83% heavier than TLBO Mean design, respectively.

A sensitivity analysis is done on earlier mentioned parameters' variation based on the best results of a series of 101 runs. The low-cost objective function design sensitivity analysis achievements are gathered in **Tables III** and **IV**.

Table III proposes considerable divergence in final cost values in the variation of D_f from 300 mm to 3000 mm. Moreover, it is obvious that using the stronger concrete, from 20 MPa to 50 MPa, can be ended up to a cheaper design through using a concrete with a compressive strength of 55 MPa makes no improvement respect to 50 MPa. In this case but, WOA is the most sensitive algorithm to the variation of f_c and FA is the least one. Changing force inclination respect to the vertical direction causes diverse observations. Referring **Table III**, by enhancing the amount of inclination the final cost will increase. In this case, TLBO with variation percentage of -44.4% is the least sensitive and LKH with -200.92% is the most sensitive approach, respectively. Form **Table III**, it can be observed that by enhancing i from 0° to 20° the variations are minor, though after 25° these changes are considerably high.

Table IV, also demonstrates that this case study is neutral to the variation of other input parameters (i.e., ϕ , γ , E_s , and ν). The results propose neither increase/decrease nor invariant stream.

C. Case III: Axial and Flexural Loading With Dynamic Location

In this numerical simulation, Case II is resolved with two additional decision variables. In this way, the location of the column is not fixed and can be movable within the two-dimensional space at the top of the foundation. Therefore, the eccentricity of the column along with X- and Y-directions considered as two spare design variables. The most important advantage of this additive is transforming the pressure distribution under the footing toward a uniform one. This fact can prevent the foundation structure from having an unbalanced strength. The collected results in **Tables I** and **II** show that LKH finds the lowest cost design of \$48,720.24 for this problem. However, TLBO gained the lowest values of mean and SD accordingly: \$56,061.12 and \$5078.10. In this case, WOA performed the weakest record with the highest values of best, worst, mean, and SD of \$62,875.04, \$726,015.73, \$147,202.86, and \$96,514.92, respectively. Among all of the utilized algorithms, in this case, TLBO expresses the best performance by the lowest value of Worst, Mean, and STD cost design of \$80344.96, \$56061.12, \$5078.10, respectively.

Furthermore, the final results indicate the positive influence of adding two mentioned design variables on the final cost level.

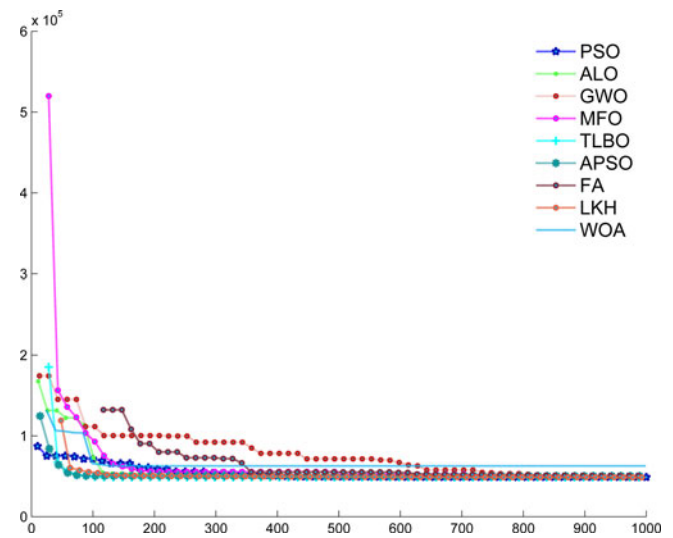


Fig. 6. Convergence rate plot for the best-found solutions for case III.

In this way, it can be observed that adding two design variables, helped APSO, FA, LKH, WOA, ALO, GWO, MFO, and TLBO to reduce the final Best design 39.26%, 39.36%, 40.99%, 27.17%, 36.59%, 40.64%, 40.48%, and 40.04%, respectively. Considering the column position dynamically, lessen the final cost considerably (nearly 40% in most of the cases) which the design led toward saving a huge amount of money.

The convergence rate plot in **Fig. 6** is a clear evidence of a fast convergence of APSO, LKH, and TLBO rather than other utilized methods.

VI. CONCLUSION

The present study is devoted to the exploring of eight meta-heuristic optimization algorithms' performance (APSO, FA, LKH, WOA, ALO, GWO, MFO, and TLBO) on reinforced concrete shallow footing optimization. In order to examine the performance of the mentioned algorithms, a MATLAB code is developed based on ACI 318-05 [14] regulations. For the numerical modeling, three case studies are suggested as follows: the first for only a uniaxial loading case, the second is a combination of a uniaxial force and a flexural moment, and finally, the third case resolved the second case by considering two additional design variables. In the third case, the position of the column is considered to change dynamically within the search procedure. It can be seen clearly that adding a flexural moment to the footing causes about 90% heavier designs. On the other hand, considering a movable situation for the column helps to save an average of 37.8% of the final cost expenses. All the algorithms have been run in a series of 101 times and the results reported in the form of best, worst, mean, SD, and median. In Case II and III, TLBO algorithm was successful to find the lowest Mean and SD values while in the Case I, GWO found the most inexpensive result. However, TLBO in Case I found a Mean result nearly close to GWO. Moreover, Best results records state that TLBO found the lowest cost value in both Cases I and II. But in Case III TLBO found a nearly close cost design to the minimum solution. To sum up, the best performance in this study observed

by TLBO because of more satisfactory results over all the case studies.

Along with the routine optimization of the mentioned case studies, a sensitivity analysis of the proposed problems is conducted for the input variables' variations accordingly: base soil friction angle variation between 28° and 38° , soil density between 11.5 and 23.5 kN/m³, modulus of elasticity between 10,500 and 90,500 kPa, Poisson's ratio between 0.1 and 1, concrete compressive strength between 20 and 55 MPa, effective force inclination respect to vertical axis between 0° and 45° , and depth of footing between 300 mm and 3000 mm. Variation of D_f , f_c , and i impacted the final cost considerably. Based on the results, increasing the depth of footing from 300 mm to 1000 mm in the Case II, causes the cheaper design though footing in a deeper level makes no improvement. Using stronger concrete also resulted in the more efficient design. However, enhancing the effective force inclination respect to the vertical direction is equivalent to a heavier design in all the cases. Other mentioned input parameters demonstrated an unimportant effect on the final design.

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